

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2083

Koch Oil 100 kV
May 22, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2083. The Interconnection Customer proposes a 203.7 MW battery energy storage facility interconnecting at the Koch Oil 100 kV substation in the Ohio area. The request was received on May 12, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 698 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,874,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gould-Ford 100kV line	Tracy-Fort Hall 100kV	121.1	148	25.2
Tracy-Gould 100kV Cir. 1	Tracy Substation-Gould 100kV line	107.3	155	19.8
Gould-Tracy 100kV Ckt 2	Gould-Tracy 100kV Line 1	107.3	155	19.8
Edic-Allen Junction 100kV	Minnehaha-Sta S Van Nuys 230kV L	106.8	121	5.5
Dower-Japanese Mesa 100kV Line	Japanese Mesa Substation-Dower Substation 100kV Cir. 1	103.2	184	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gould-Ford 100kV	Rebuild line with higher-capacity conductor	\$1,951,000	58% (\$1,137,000)
Tracy-Gould 100kV Cir. 1	Reconductor line	\$1,337,000	60% (\$797,000)
Gould-Tracy 100kV	Reconductor line	\$1,357,000	60% (\$809,000)
Edic-Allen Jct Substation 100kV	Reconductor line	\$1,652,000	50% (\$817,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Japanese Mesa Substation-Dower 100kV Line 2	Reconductor line	\$368,000	86% (\$314,000)
	Total allocated to project		\$3,874,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.